

# Notes: Koijen, Richmond, & Yogo (RES, 2024)

## Which Investors Matter for Equity Valuations and Expected Returns?

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# 1 Big picture

- **Question.** Which investors move equity prices, expected returns, price informativeness, and investor wealth when capital flows or preferences change?
- **Core object.** The paper estimates an investor-level asset demand system: for each investor, how portfolio weights respond to stock prices and firm characteristics.
- **Data.** U.S. institutional holdings from FactSet Ownership, 2000.Q1–2019.Q4, combined with firm data from FactSet/Compustat and characteristics such as environmental score, governance, size, foreign sales, market power, payout, and beta.
- **Method.** Estimate heterogeneous investor demand curves with an instrumental-variables ridge estimator, then solve market-clearing counterfactuals.
- **Application 1: active to passive.** The decline in aggregate active share mostly reflects capital flows from active to passive investors, not investors becoming more passive. The counterfactual generates large equity repricing, about 14%, but little change in price informativeness.
- **Application 2: climate demand shifts.** A broad shift toward greener stocks can have a large price and wealth-distribution effect. Passive advisors, long-term investors, and private banking gain; active advisors and hedge funds lose.
- **Investor importance.** Size matters, but so do demand heterogeneity and price elasticity. Hedge funds have relatively small wealth share but large price impact per dollar because their portfolios differ strongly and their demand is relatively elastic.
- **Takeaway.** Asset prices are not moved by “the representative investor.” They are equilibrium objects shaped by which investors own wealth, which characteristics each investor values, and how elastic each investor’s demand is.

## 2 Data and measurement

### 2.1 Investor groups

The paper groups investors into households, investment advisors, hedge funds, long-term investors, private banking, brokers, and foreign investors. Investment advisors are further split by size and active share into large-passive, small-passive, large-active, and small-active investment advisors.

**Interpretation.** This classification matters because the paper is not only estimating aggregate demand. It asks which investor groups are influential. The taxonomy therefore links micro holdings to counterfactual market outcomes.

### 2.2 Firm universe and characteristics

The inside assets are the largest 90% of U.S. firms by market equity. The remaining 10% are grouped into an outside asset. The baseline demand characteristics are:

- environmental score and governance index, available mainly for 2010–19;
- log book equity, foreign sales share, Lerner index, sales-to-book, dividends-to-book, and market beta, available for 2000–19.

**Interpretation.** The largest 90% of firms capture most equity value and economic activity, while the outside asset prevents the model from forcing investors to allocate only to the estimated inside stocks. Environmental and governance data are shorter, so the paper uses two estimation samples.

### 2.3 Key measurements

- **Market-to-book equity.** The valuation object explained by firm characteristics and investor demand.
- **Active share.** One-half the sum of absolute deviations between an investor’s portfolio weights and market weights over that investor’s investment universe.

- **Equity repricing.** The value-weighted absolute percentage change in prices between actual and counterfactual equilibria.
- **Price informativeness.** The coefficient from a cross-sectional regression of future profitability on valuation, following Bai et al. (2016).
- **Investor-level informativeness.** The predictive content of an investor's demand shifters for future profitability.

**Interpretation.** Active share measures how different an investor is from the market. Repricing measures the size of equilibrium price changes. Informativeness asks whether prices forecast future profitability, not whether they merely move a lot.

### 3 Model and empirical demand system

#### 3.1 Stylized model: demand and equilibrium prices

Investors choose stock holdings in units of book equity,  $Q_i(n)$ . The stylized CARA-normal model implies a linear demand curve:

$$Q_i(n) = \frac{1}{\gamma_i \sigma^2} [-MB(n) + \beta'_i x(n) + \epsilon_i(n)].$$

Here  $MB(n)$  is market-to-book equity,  $x(n)$  is the vector of observed firm characteristics,  $\beta_i$  is investor  $i$ 's demand loading on those characteristics, and  $\epsilon_i(n)$  is latent demand.

Market clearing implies:

$$MB(n) = \bar{\beta}' x(n) + \bar{\epsilon}(n), \quad \bar{\beta} = \sum_i a_i \beta_i, \quad \bar{\epsilon}(n) = \sum_i a_i \epsilon_i(n).$$

The weights  $a_i$  depend on wealth and risk tolerance.

**Interpretation.** Equity values are weighted averages of investor demand. Capital flows matter because they change the weights  $a_i$ . Preference or regulatory shifts matter because they change the demand loadings  $\beta_i$ .

#### 3.2 Logit demand used for estimation

For estimation, the paper uses a logit demand system because holdings are closer to lognormal than linear. Investor  $i$ 's portfolio weight on stock  $n$  is:

$$w_{i,t}(n) = \frac{\delta_{i,t}(n)}{1 + \sum_{m \in \mathcal{N}_{i,t}} \delta_{i,t}(m)},$$

where

$$\delta_{i,t}(n) = \exp(\alpha_{i,t} + \beta_{0,i,t} mb_t(n) + \beta'_{1,i,t} x_t(n)) \epsilon_{i,t}(n).$$

The outside-asset weight is:

$$w_{i,t}(0) = \frac{1}{1 + \sum_{m \in \mathcal{N}_{i,t}} \delta_{i,t}(m)}.$$

**Interpretation.** A lower coefficient on log market-to-book equity,  $\beta_{0,i,t}$ , means more price-elastic demand. The latent-demand term captures investor-specific demand that is not explained by observable characteristics.

#### 3.3 Market clearing and counterfactual wealth

Market clearing is:

$$P_t(n) = \sum_{i=1}^I A_{i,t} w_{i,t}(n; P_t).$$

For a counterfactual price vector  $P_t^C$ , investor wealth changes endogenously:

$$A_{i,t}^C(P_t^C) = A_{i,t} \left[ w_{i,t}(0) + \sum_{n \in \mathcal{N}_{i,t}} \frac{P_t^C(n)}{P_t(n)} w_{i,t}(n) \right] + F_{i,t}.$$

Counterfactual prices solve:

$$P_t^C(n) = \sum_{i=1}^I A_{i,t}^C(P_t^C) w_{i,t}^C(n; P_t^C).$$

**Interpretation.** This is the key innovation for policy analysis. The paper does not simply shock one stock or one investor. It solves a new equilibrium with substitution across stocks and wealth changes across investors.

### 3.4 Identification and estimation

The demand estimation faces two problems.

- **Endogeneity.** Market-to-book equity and latent demand are jointly determined.
- **Small portfolios.** Many investors hold concentrated portfolios, so individual demand estimates can be noisy.

The paper instruments for log market equity using other investors' estimated investment universes. The instrument is based on the idea that investment mandates and benchmark universes shift aggregate demand for a stock, but affect a given investor's portfolio choice only through prices. The paper also uses a two-step IV ridge estimator, shrinking small investors toward group-level estimates and choosing the penalty by cross-validation.

**Interpretation.** The IV strategy turns cross-sectional variation in other investors' mandate-driven investment universes into price variation. The ridge step is important because the whole project requires investor-level heterogeneity rather than only investor-type averages.

## 4 Identification and caveats

- **Instrument assumption.** Other investors' investment universes affect investor  $i$  through stock prices, not through investor  $i$ 's unobserved demand shocks.
- **Investment universe construction.** The universe is approximated as stocks an investor currently holds or has held in the prior eleven quarters. The paper checks alternative windows and random universe expansions.
- **Characteristic exogeneity.** The baseline model treats firm characteristics other than valuation as exogenous. The paper checks robustness to treating dividend-to-book as endogenous.
- **Short-run counterfactuals.** Firm characteristics, shares outstanding, and environmental scores are held fixed. This is appropriate for short-run price effects, but not for long-run real responses.
- **No full welfare analysis.** The framework allows wealth redistribution calculations, but welfare with heterogeneous beliefs is not solved.

**Interpretation.** The paper is strongest as a quantitative equilibrium measurement exercise. It answers "what prices and wealth would clear the market under a specified demand shock," not "what is the socially optimal regulation."

## 5 Tables and figures

### 5.1 Table 1 and Figure 1: investor groups and ownership shares

TABLE 1  
Largest investors by investor type

| Type             | Investor                             | Wealth |
|------------------|--------------------------------------|--------|
| Households       |                                      | 8,553  |
| Large-passive IA | The Vanguard Group                   | 2,494  |
| Large-active IA  | T. Rowe Price Associates             | 659    |
| Long-term        | Norges Bank Investment Management    | 292    |
| Small-passive IA | Charles Schwab Investment Management | 162    |
| Small-active IA  | PRIMECAP Management Company          | 117    |
| Private banking  | Goldman Sachs and Company            | 112    |
| Hedge funds      | Renaissance Technologies             | 89     |
| Brokers          | Schweizerische Nationalbank          | 84     |
| Foreign          | Norges Bank Investment Management    | 292    |

Note: Wealth is U.S. equity holdings in billions of U.S. dollars in 2019.Q4. IA is abbreviation for investment advisors.

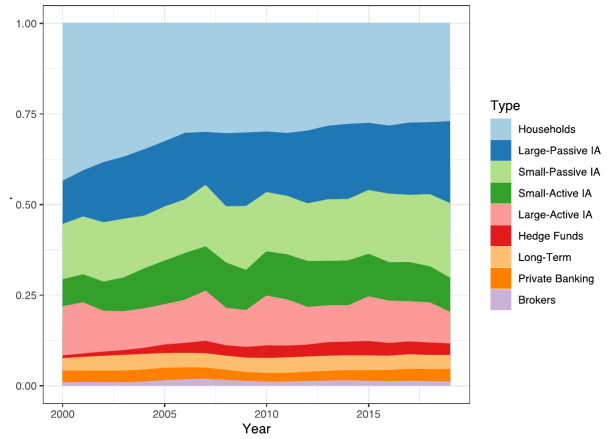


FIGURE 1  
Ownership shares by investor type  
This figure reports the ownership shares for the U.S. stock market by investor type. IA is abbreviation for investment advisors. Household share is based on shares outstanding minus institutional ownership. The quarterly sample period is 2000.Q1 to 2019.Q4.

#### Read:

- Table 1 shows the largest investor in each group. Households are the largest sector; among institutions, Vanguard is the largest large-passive advisor.
- Figure 1 shows ownership shares from 2000 to 2019. Institutional ownership rises, and passive investors become increasingly important.
- The investor classification is used throughout the counterfactuals.

**Economic message:** The demand system needs both holdings and investor identity. The figure/table pair motivates why passive investment advisors, active advisors, hedge funds, and long-term investors are economically distinct players.

### 5.2 Tables 2 and 3: firm sample and motivating valuation facts

TABLE 2  
Firm size distribution

| Market equity percentile | Number of firms | Sales percentile | Earnings percentile |
|--------------------------|-----------------|------------------|---------------------|
| Panel A: 2019.Q4         |                 |                  |                     |
| 10                       | 3               | 4                | 6                   |
| 20                       | 9               | 12               | 15                  |
| 30                       | 19              | 21               | 26                  |
| 40                       | 34              | 29               | 40                  |
| 50                       | 57              | 34               | 46                  |
| 60                       | 97              | 44               | 58                  |
| 70                       | 159             | 56               | 69                  |
| 80                       | 278             | 68               | 78                  |
| 90                       | 541             | 81               | 88                  |
| 100                      | 2,825           | 100              | 100                 |
| Panel B: 2000.Q1         |                 |                  |                     |
| 10                       | 3               | 2                | 3                   |
| 20                       | 8               | 7                | 7                   |
| 30                       | 17              | 12               | 15                  |
| 40                       | 29              | 16               | 19                  |
| 50                       | 48              | 21               | 30                  |
| 60                       | 80              | 34               | 47                  |
| 70                       | 142             | 42               | 55                  |
| 80                       | 274             | 55               | 68                  |
| 90                       | 615             | 72               | 83                  |
| 100                      | 5,137           | 100              | 100                 |

Notes: This table reports summary statistics for publicly traded U.S. firms by deciles of market equity. The last column is for earnings before interest and taxes from Compustat (S&P Global, 2020).

TABLE 3  
Explaining equity valuations and future profitability with firm characteristics

| Characteristic        | 2010–19           |                   | 2000–19           |                   |
|-----------------------|-------------------|-------------------|-------------------|-------------------|
|                       | Market-to-book    | Profitability     | Market-to-book    | Profitability     |
| Environment           | 0.17<br>(5.68)    | 0.06<br>(2.49)    |                   |                   |
| Governance            | −0.10<br>(−3.79)  | −0.08<br>(−3.67)  |                   |                   |
| Log book equity       | −0.65<br>(−23.89) | −0.23<br>(−12.05) | −0.55<br>(−25.55) | −0.20<br>(−14.00) |
| Foreign sales         | 0.11<br>(5.17)    | 0.01<br>(0.79)    | 0.14<br>(8.48)    | 0.04<br>(3.45)    |
| Lerner                | 0.08<br>(3.14)    | 0.14<br>(7.11)    | 0.09<br>(4.02)    | 0.16<br>(11.21)   |
| Sales to book         | 0.22<br>(8.12)    | 0.20<br>(9.31)    | 0.21<br>(10.21)   | 0.18<br>(8.97)    |
| Dividends to book     | 0.17<br>(8.18)    | 0.09<br>(5.64)    | 0.19<br>(11.00)   | 0.07<br>(5.61)    |
| Market beta           | −0.04<br>(−2.40)  | −0.03<br>(−2.53)  | 0.02<br>(1.40)    | −0.00<br>(−0.42)  |
| Adjusted $R^2$        | 0.65              | 0.50              | 0.57              | 0.38              |
| Adjusted within $R^2$ | 0.64              | 0.49              | 0.55              | 0.37              |
| Observations          | 6,399             | 2,102             | 13,664            | 6,576             |

Notes: This table reports regressions of log market-to-book equity or profitability on firm characteristics. Log market-to-book equity is measured at the end of year  $t$ . Profitability is the sum of discounted future clean-surplus earnings from year  $t$  to  $t+5$ . All characteristics are measured in year  $t$  and are standardized within each year. The environment scores are from Sustainalytics (Morningstar, 2020). The governance index is the number of entrenchment provision following Bebchuk et al. (2009). The environmental scores and the governance indices are available from 2010 to 2015. The specifications with these variables include indicator variables for a missing environmental score or governance index. The Lerner index is the ratio of operating income after depreciation to sales. All specifications include year fixed effects. The  $t$ -statistics, based on robust standard errors clustered by firm, are reported in parentheses.

#### Read:

- Table 2 shows strong market concentration: in 2019.Q4, the largest 57 firms account for 50% of market equity, while the largest 541 firms account for 90%.
- The largest 90% of firms by market equity also represent most sales and earnings, justifying their use as inside assets.
- Table 3 shows that firm characteristics explain a large share of market-to-book equity and also predict future profitability.
- In 2010–19, the adjusted  $R^2$  is 0.65 for market-to-book and 0.50 for future profitability.

**Economic message:** Firm characteristics are not arbitrary controls. They are both valuation characteristics and signals about future profitability, so investor demand for those characteristics has asset-pricing content.

### 5.3 Figure 2 and Table 4: heterogeneous demand across investors

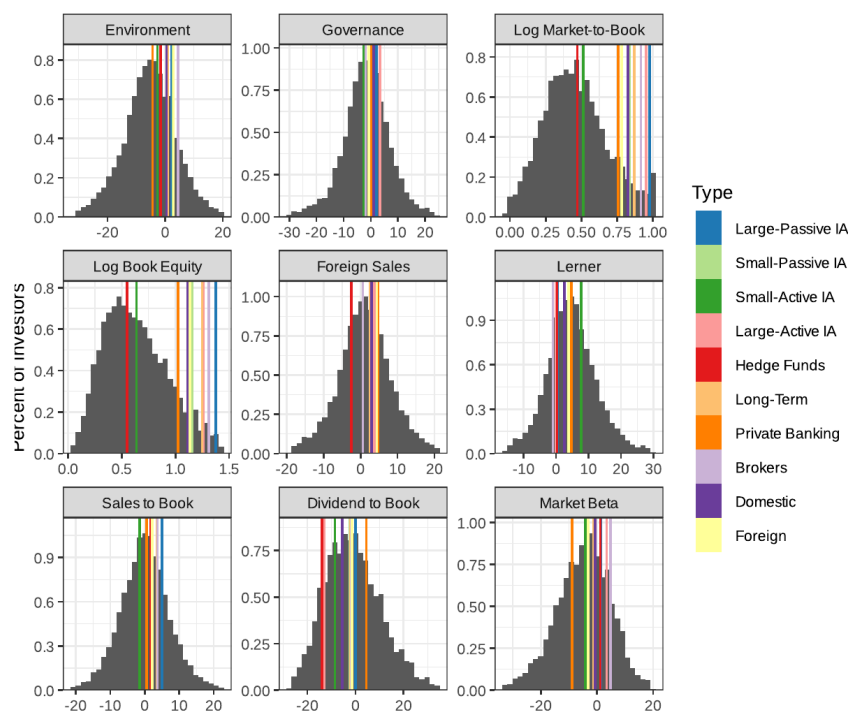


FIGURE 2

Heterogeneity in asset demand by investor type

This figure reports the cross-sectional distribution of the average demand coefficients across investors. For each investor, we estimate asset demand on quarterly holdings pooled by year with quarter fixed effects. We then compute the time-series average of the estimated demand coefficients for each investor over the sample period during which both investor holdings and firm characteristics are available (limited to 2010–19 for the environmental score and the governance index). Finally, we compute a wealth-weighted average demand coefficient by investor type, in which an investor's weight is the time-series average of its wealth share by investor type (an investor's wealth is its total equity holdings). The coloured vertical lines represent the wealth-weighted average demand coefficient by investor type. IA is abbreviation for investment advisors. The environmental scores are from Sustainalytics ([Morningstar, 2020](#)). The governance index is the number of entrenchment provisions, following [Bebchuk et al. \(2009\)](#). The Lerner index is the ratio of operating income after depreciation to sales. We multiply the demand coefficients by 100, except those for log market-to-book equity and log book equity.

TABLE 4  
Heterogeneity in asset demand by investor type, wealth, and active share

| Investor characteristic                                    | Environment      | Governance       | Log market-to-book | Log book equity  | Foreign sales    | Lerner           | Sales to book    | Dividends to book  | Market beta      |
|------------------------------------------------------------|------------------|------------------|--------------------|------------------|------------------|------------------|------------------|--------------------|------------------|
| <i>Panel A: investor type</i>                              |                  |                  |                    |                  |                  |                  |                  |                    |                  |
| Hedge funds                                                | -1.56<br>(-2.35) | 0.70<br>(1.59)   | 0.47<br>(27.15)    | 0.55<br>(25.74)  | -2.51<br>(-8.91) | 0.29<br>(0.79)   | 1.81<br>(2.96)   | -13.94<br>(-26.20) | 1.20<br>(2.18)   |
| Investment advisors:                                       |                  |                  |                    |                  |                  |                  |                  |                    |                  |
| Large-passive                                              | 2.25<br>(2.61)   | 2.10<br>(2.35)   | 0.98<br>(60.11)    | 1.38<br>(48.35)  | 3.78<br>(4.11)   | 0.49<br>(1.13)   | 5.00<br>(5.09)   | 0.01<br>(0.00)     | 1.46<br>(1.55)   |
| Small-passive                                              | 2.82<br>(4.73)   | 0.89<br>(1.12)   | 0.83<br>(56.32)    | 1.16<br>(73.16)  | 3.12<br>(8.09)   | 3.83<br>(8.62)   | 1.81<br>(2.79)   | -2.27<br>(-2.54)   | -3.52<br>(-6.80) |
| Small-active                                               | -2.71<br>(-3.23) | -2.71<br>(-3.98) | 0.51<br>(27.95)    | 0.64<br>(27.86)  | 2.80<br>(5.97)   | 7.68<br>(11.57)  | -1.48<br>(-2.30) | -8.59<br>(-6.87)   | -4.17<br>(-5.16) |
| Large-active                                               | 0.72<br>(0.74)   | 3.36<br>(3.08)   | 0.95<br>(58.61)    | 1.26<br>(43.28)  | 3.77<br>(3.54)   | 0.02<br>(0.01)   | 2.17<br>(1.42)   | -13.15<br>(-5.21)  | 3.43<br>(2.53)   |
| Long-term                                                  | 0.94<br>(1.48)   | -0.13<br>(-0.25) | 0.87<br>(51.06)    | 1.25<br>(70.57)  | 2.52<br>(6.00)   | 3.80<br>(7.92)   | 3.53<br>(6.45)   | -2.08<br>(-3.08)   | -1.22<br>(-2.13) |
| Private banking                                            | -4.39<br>(-2.18) | 0.22<br>(0.24)   | 0.76<br>(23.25)    | 1.02<br>(15.75)  | 4.61<br>(6.69)   | 4.74<br>(3.69)   | 0.57<br>(0.48)   | 4.49<br>(2.01)     | -8.85<br>(-5.19) |
| Brokers                                                    | 4.40<br>(4.68)   | -1.87<br>(-2.56) | 0.92<br>(47.09)    | 1.31<br>(51.51)  | 0.59<br>(0.81)   | -0.98<br>(-0.96) | 3.66<br>(3.96)   | -1.64<br>(-1.01)   | 4.88<br>(2.27)   |
| Adjusted $R^2$                                             | 0.08             | 0.07             | 0.49               | 0.59             | 0.05             | 0.15             | 0.07             | 0.16               | 0.15             |
| Observations                                               | 6,560            | 6,560            | 7,959              | 7,959            | 7,959            | 7,959            | 7,959            | 7,959              | 7,959            |
| <i>Panel B: wealth, active share, and foreign investor</i> |                  |                  |                    |                  |                  |                  |                  |                    |                  |
| Log wealth share                                           | 0.94<br>(3.38)   | 1.24<br>(3.59)   | 0.11<br>(18.72)    | 0.13<br>(11.28)  | 0.59<br>(2.19)   | -2.01<br>(-6.50) | 0.21<br>(0.56)   | -2.84<br>(-2.51)   | 2.47<br>(6.62)   |
| Active share                                               | -1.37<br>(-3.91) | -0.14<br>(-0.49) | -0.07<br>(-7.55)   | -0.15<br>(-9.71) | -0.06<br>(-0.24) | 0.18<br>(0.61)   | -1.51<br>(-4.15) | -4.72<br>(-8.84)   | 0.37<br>(1.05)   |
| Foreign                                                    | 3.00<br>(3.81)   | -0.79<br>(-1.40) | 0.04<br>(2.56)     | 0.10<br>(4.98)   | 1.92<br>(3.91)   | -0.31<br>(-0.52) | -0.12<br>(-0.19) | 0.88<br>(0.97)     | -0.38<br>(-0.53) |
| Adjusted $R^2$                                             | 0.11             | 0.04             | 0.56               | 0.67             | 0.02             | 0.12             | 0.06             | 0.13               | 0.09             |
| Observations                                               | 6,560            | 6,560            | 7,959              | 7,959            | 7,959            | 7,959            | 7,959            | 7,959              | 7,959            |

Notes: This table reports regressions of the average demand coefficients on investor characteristics. For each investor, we estimate asset demand on quarterly holdings pooled by year with quarter fixed effects. The environmental scores are from Sustainalytics (Morningstar, 2020). The governance index is the number of entrenchment provisions, following Bebchuk *et al.* (2009). The Lerner index is the ratio of operating income after depreciation to sales. For each investor, the average demand coefficients are time-series averages over the sample period over which both investor holdings and firm characteristics are available (limited to 2010–19 for the environmental score and the governance index). We multiply the demand coefficients by 100, except those for log market-to-book equity and log book equity. The investor characteristics in panel A are investor-type fixed effects. The investor characteristics in panel B are the log average wealth share, the active share, and a foreign-investor fixed effect. An investor's wealth is its total equity holdings. The log average wealth share and the active share are standardized. The regression weights the observations by the time-series average of the investor's wealth share. The  $t$ -statistics, based on heteroskedasticity-robust standard errors, are reported in parentheses.

## Read:

- Figure 2 shows substantial cross-investor dispersion in demand coefficients, even within investor types.
- Small-passive investment advisors have a wealth-weighted environmental coefficient of 2.82%; hedge funds have -1.56%.
- Price elasticity is low on average. Even hedge funds, the most elastic institutional group, have wealth-weighted price elasticity around 0.5.
- Table 4 confirms that investor type explains some demand heterogeneity, but not all of it. Log market-to-book and log book equity loadings are strongly related to investor characteristics.

**Economic message:** The central empirical input is heterogeneity. Without heterogeneous demand slopes and latent demand, the counterfactuals would collapse into a representative-investor calculation.

## 5.4 Figures 3–4: foreign versus domestic demand and the active-share trend

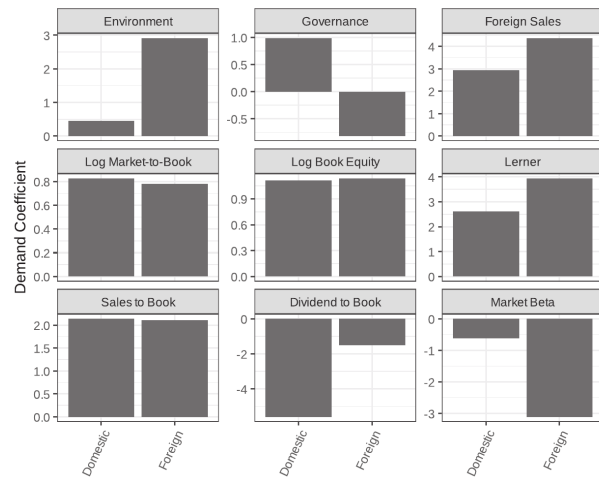


FIGURE 3

Heterogeneity in asset demand between domestic and foreign investors

*Text:* This figure reports the wealth-weighted average demand coefficients for domestic and foreign investors. An investor's wealth is its total equity holdings. For each investor, we estimate asset demand on quarterly holdings pooled by year with quarter fixed effects. We then compute the time-series average of the estimated demand coefficients for each investor over the sample period during which both investor holdings and firm characteristics are available (limited to 2010–19 for the environmental score and the governance index). Finally, we compute a wealth-weighted average of the demand coefficients by investor type, in which an investor's weight is the time-series average of its wealth share by investor type. The environmental scores are from Sustainalytics (Morningstar, 2020). The governance index is the number of entrenchment provisions, following Bebchuk *et al.* (2009). The Lerner index is the ratio of operating income after depreciation to sales. We multiply the demand coefficients by 100, except those for log market-to-book equity and log book equity.

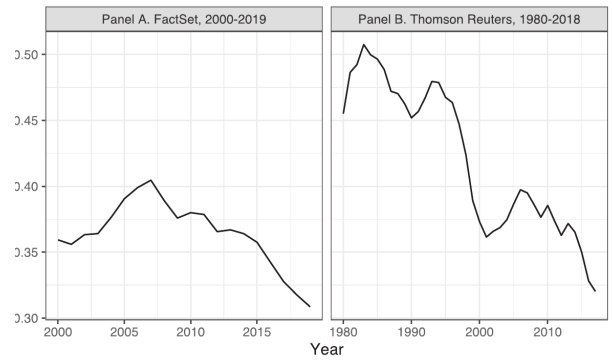


FIGURE 4

Aggregate active share

For each investor, the active share is one-half times the sum of the absolute differences between the portfolio weights and the weights over the set of stocks that are in its investment universe. The aggregate active share is a wealth-weighted average of the active shares across all institutional investors, in which an investor's weight is its wealth as a share of total wealth. An investor's wealth is its equity holdings. This figure reports an annual average of the quarterly estimates of the aggregate active share. Panel A is based on Ownership from 2000.Q1 to 2019.Q4 (FactSet, 2020). Panel B is based on the Thomson Reuters Institutional Holdings Database from 1980.Q1 to 2018.Q4 (Koijen and Yogo, 2019).

### Read:

- Figure 3 compares domestic and foreign investors. Foreign investors have higher loadings on environment, foreign sales, Lerner, and a more negative market-beta loading.
- Figure 4 documents the active-to-passive trend. In the paper's main FactSet window, aggregate active share declines from 38.9% in 2007.Q4 to 32.8% in 2016.Q4.
- In the longer Thomson Reuters series, active share falls from 46.6% in 1980.Q4 to 31.7% in 2017.Q4.
- The 2007–16 active-share change decomposes into -1.3% from strategy changes and -4.8% from wealth-distribution changes.

**Economic message:** The active-to-passive shift is mainly a capital-flow shock. Investors did not simply make the same portfolios more passive; passive investors became more important owners.



## 5.5 Figure 5: counterfactual market under the 2007 wealth distribution

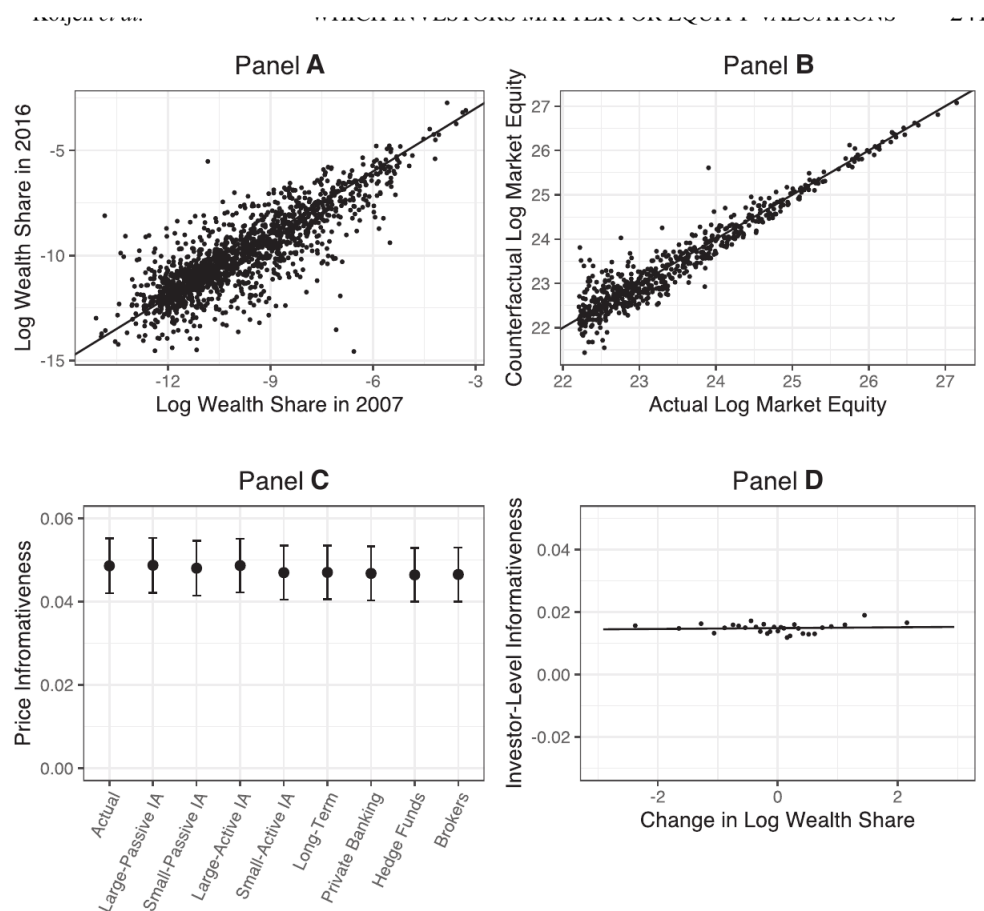


FIGURE 5

Counterfactual market in 2016 under the 2007 wealth distribution

Notes: Panel A is a scatter plot of the log wealth shares in 2007.Q4 versus 2016.Q4. An investor's wealth is its total equity holdings. Panel B is a scatter plot of the actual log market equity in 2016.Q4 versus the counterfactual log market equity under the wealth distribution in 2007.Q4. In the counterfactual, wealth changes only for institutional investors who exist in both periods. Panel C reports the cumulative effects on price informativeness when we change the wealth distribution by sequentially adding each investor type. The range around the joint estimate represents the 95% confidence interval. Panel D is a bin scatter plot of the changes in the log wealth share from 2007.Q4 to 2016.Q4 versus the investor-level informativeness in 2016.Q4.

### Read:

- Panel A shows investor wealth shares changed substantially between 2007.Q4 and 2016.Q4.
- Panel B shows counterfactual equity prices in 2016 under the 2007 wealth distribution. Smaller stocks are especially affected.
- The paper summarizes the price movement with equity repricing: value-weighted absolute price change is about 14%.
- Panel C shows that price informativeness barely changes as investor groups are added to the counterfactual.
- Panel D shows little relation between investors' wealth-share changes and investor-level informativeness.

**Economic message:** Passive investing has a large valuation effect but not a large informativeness effect in this paper. The reason is that capital did not flow systematically from more informed to less informed investors.

## 5.6 Table 5 and Figure 6: climate-induced demand shifts

TABLE 5  
Impact of climate risk on valuation regressions

| Characteristic    | Actual            | Counterfactual   |                  |
|-------------------|-------------------|------------------|------------------|
|                   |                   | Stakeholder      | Regulatory       |
| Environment       | 0.23<br>(4.19)    | 0.57<br>(47.06)  | 0.03<br>(30.26)  |
| Governance        | -0.14<br>(-2.24)  | -0.01<br>(-0.43) | -0.00<br>(-1.83) |
| Log book equity   | -0.74<br>(-16.39) | -0.01<br>(-1.02) | -0.00<br>(-1.77) |
| Foreign sales     | 0.11<br>(3.31)    | 0.00<br>(0.31)   | -0.00<br>(-0.40) |
| Lerner            | 0.11<br>(2.68)    | 0.02<br>(1.54)   | -0.00<br>(-0.34) |
| Sales to book     | 0.22<br>(4.63)    | -0.00<br>(-0.41) | -0.00<br>(-2.03) |
| Dividends to book | 0.16<br>(4.17)    | 0.01<br>(1.03)   | 0.00<br>(2.11)   |
| Market beta       | -0.04<br>(-1.26)  | 0.00<br>(0.67)   | 0.00<br>(1.81)   |
| Adjusted $R^2$    | 0.65              | 0.92             | 0.81             |
| Observations      | 540               | 540              | 540              |

Notes: The first column is a regression of log market-to-book equity on firm characteristics in 2019:Q4. The second and third columns are regressions of the difference between actual and counterfactual log market-to-book equity on firm characteristics. All characteristics are standardized within each year. For stakeholder risk, asset demand for a institutional investors shifts as the coefficient on the environmental score increases by 0.1. For regulatory risk, asset demand for only long-term investors shifts as the coefficient on the environmental score increases by 0.1. All specifications include indicator variables for a missing environmental score or governance index. The  $t$ -statistics, based on heteroskedasticity-robust standard errors, are reported in parentheses.

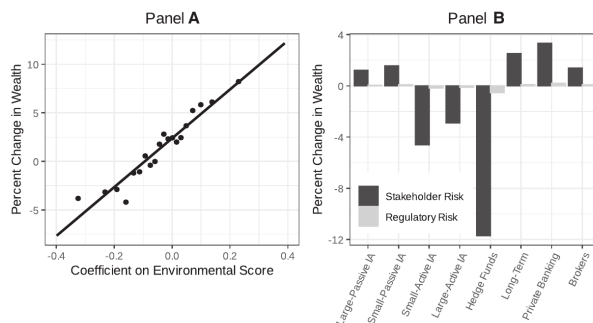


FIGURE 6  
Impact of climate risk on the wealth distribution across institutional investors  
Notes: Panel A is a bin scatter plot of the demand coefficient on the environmental score versus the percent change in log wealth in response to realized stakeholder risk. An investor's wealth is its total equity holdings. Panel B reports the percent change in log wealth by investor type in response to a climate-induced shift in asset demand. For each investor type, we compute a wealth-weighted average of the log change in wealth, in which an investor's weight is its wealth share by investor type. IA is abbreviation for investment advisor. Stakeholder risk, asset demand for all institutional investors shifts as the coefficient on the environmental score increases by 0.1. For regulatory risk, asset demand for only long-term investors shifts as the coefficient on the environmental score increases by 0.1. Each sorted value is a time-series average of the quarterly estimates from 2019:Q1 to 2019:Q4.

### Read:

- The stakeholder-risk counterfactual increases every institutional investor's environmental demand coefficient by 0.1.
- The regulatory-risk counterfactual increases only long-term investors' environmental demand coefficient by 0.1.
- Table 5 shows that stakeholder risk raises the valuation coefficient on environment by 0.57, while regulatory risk raises it only by 0.03.
- Figure 6 shows that investors initially tilted toward greener stocks gain when green stocks become more expensive.
- Passive advisors, long-term investors, and private banking gain; active advisors and hedge funds lose.

**Economic message:** A broad sustainable-investing shift can materially reprice green versus brown firms and redistribute wealth across investors. A narrower regulatory constraint on pension funds and insurers has a smaller equity-market effect.

## 5.7 Table 6 and Figure 7: which investor types matter for prices?

TABLE 6  
Equity repricing for counterfactual outflows by investor type

| Investor type        | Wealth share (%) | Repricing | Repricing per dollar wealth |
|----------------------|------------------|-----------|-----------------------------|
| Investment advisors: |                  |           |                             |
| Large-passive        | 17.7             | 15.9      | 0.90                        |
| Small-passive        | 16.4             | 17.2      | 1.05                        |
| Small-active         | 11.7             | 26.7      | 2.28                        |
| Large-active         | 11.1             | 18.4      | 1.65                        |
| Hedge funds          | 3.2              | 11.5      | 3.58                        |
| Long-term            | 3.9              | 3.9       | 1.01                        |
| Private banking      | 2.9              | 5.3       | 1.81                        |
| Brokers              | 1.1              | 1.8       | 1.56                        |
| Foreign              | 6.1              | 8.0       | 1.31                        |

Notes: The first column reports the wealth share by investor type. An investor's wealth is its total equity holdings. The second column reports the equity repricing for capital flows from the given group of investors to other institutional investors. Equity repricing is the value-weighted absolute percent change in equity prices. The last column reports the ratio of equity repricing in the second column to the wealth share in the first column. Each cell is a time-series average of the quarterly estimates from 2000:Q1 to 2019:Q4.

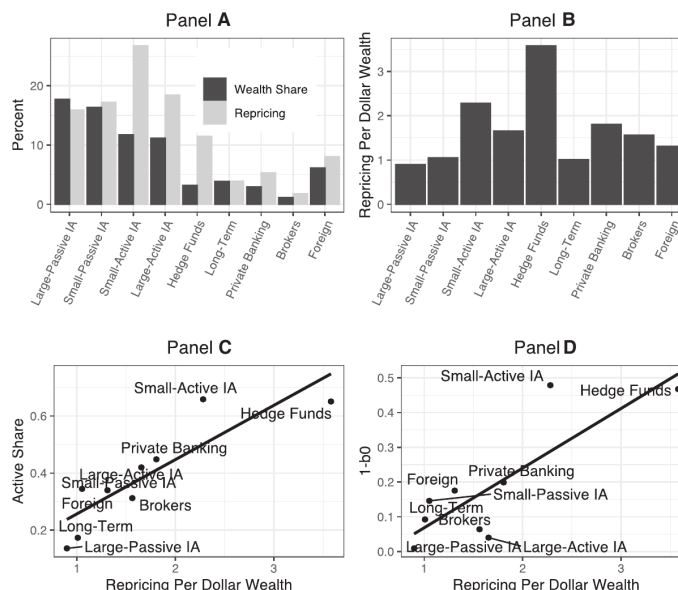


FIGURE 7

Equity repricing for counterfactual outflows by investor type

es: Panel A reports the wealth share by investor type and the equity repricing for capital flows from the given group of investor ther institutional investors. An investor's wealth is its total equity holdings. Equity repricing is the value-weighted absolute percent age in equity prices. IA is abbreviation for investment advisors. Panel B reports the equity repricing per dollar of wealth, which is th ) of equity repricing to the wealth share. Panel C is a scatter plot of the equity repricing per dollar of wealth versus the active shau investor type. For each investor, the active share is one-half times the sum of the absolute differences between the portfolio weighi the market weights over the set of stocks that are in its investment universe. Panel D is a scatter plot of the equity repricing per doll

### Read:

- Table 6 reports counterfactual outflows from each investor type to other institutional investors.
- Small-active advisors generate the largest total repricing, 26.7%, because they combine meaningful wealth share with distinctive demand.
- Hedge funds generate 11.5% total repricing despite only a 3.2% wealth share.
- Per dollar of wealth, hedge funds have the largest price impact: 3.58 dollars of repricing per dollar of wealth.
- Figure 7 shows the mechanism: price impact per dollar rises with active share and with demand elasticity.

**Economic message:** Size alone is not enough. An investor type matters when it is large, different from other investors, and price elastic enough that its withdrawal must be absorbed by relatively inelastic investors.

## 5.8 Table 7: valuation regressions under counterfactual outflows

TABLE 7  
Valuation regressions for counterfactual outflows by investor type

| Characteristic        | Actual            | Investment advisors |                   |                   |                   | Hedge funds       | Long-term         | Private banking   | Brokers           | Foreign           |
|-----------------------|-------------------|---------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                       |                   | Large-passive       | Small-passive     | Small-active      | Large-active      |                   |                   |                   |                   |                   |
| Environment           | 0.17<br>(5.68)    | 0.17<br>(5.18)      | 0.14<br>(4.44)    | 0.21<br>(6.23)    | 0.16<br>(5.39)    | 0.18<br>(5.60)    | 0.17<br>(5.54)    | 0.17<br>(5.77)    | 0.17<br>(5.60)    | 0.14<br>(4.55)    |
| Governance            | -0.10<br>(-3.79)  | -0.11<br>(-3.64)    | -0.11<br>(-3.55)  | -0.09<br>(-2.65)  | -0.12<br>(-4.79)  | -0.09<br>(-3.09)  | -0.10<br>(-3.52)  | -0.10<br>(-3.69)  | -0.10<br>(-3.70)  | -0.10<br>(-3.43)  |
| Log book equity       | -0.65<br>(-23.89) | -0.69<br>(-25.44)   | -0.73<br>(-24.92) | -0.44<br>(-14.38) | -0.66<br>(-24.44) | -0.59<br>(-20.51) | -0.67<br>(-24.45) | -0.66<br>(-24.44) | -0.65<br>(-24.01) | -0.69<br>(-24.42) |
| Foreign sales         | 0.11<br>(5.17)    | 0.13<br>(5.60)      | 0.11<br>(4.89)    | 0.07<br>(2.86)    | 0.10<br>(5.20)    | 0.12<br>(5.30)    | 0.10<br>(4.71)    | 0.11<br>(5.07)    | 0.11<br>(5.19)    | 0.09<br>(4.17)    |
| Lerner                | 0.08<br>(3.14)    | 0.08<br>(3.27)      | 0.05<br>(1.92)    | 0.05<br>(1.57)    | 0.09<br>(4.42)    | 0.09<br>(3.55)    | 0.07<br>(2.83)    | 0.07<br>(2.79)    | 0.08<br>(3.21)    | 0.06<br>(2.69)    |
| Sales to book         | 0.22<br>(8.12)    | 0.21<br>(7.66)      | 0.21<br>(7.52)    | 0.26<br>(8.09)    | 0.22<br>(8.66)    | 0.20<br>(7.08)    | 0.21<br>(7.64)    | 0.21<br>(8.14)    | 0.21<br>(8.05)    | 0.21<br>(7.48)    |
| Dividends to book     | 0.17<br>(8.18)    | 0.12<br>(5.73)      | 0.10<br>(4.33)    | 0.27<br>(10.72)   | 0.19<br>(9.51)    | 0.23<br>(10.77)   | 0.15<br>(7.23)    | 0.14<br>(7.04)    | 0.17<br>(8.11)    | 0.14<br>(6.59)    |
| Market beta           | -0.04<br>(-2.40)  | -0.03<br>(-1.41)    | -0.03<br>(-1.58)  | -0.05<br>(-2.74)  | -0.04<br>(-2.58)  | -0.06<br>(-3.09)  | -0.04<br>(-2.20)  | -0.04<br>(-2.14)  | -0.04<br>(-2.51)  | -0.03<br>(-1.56)  |
| Adjusted within $R^2$ | 0.64              | 0.61                | 0.61              | 0.45              | 0.65              | 0.57              | 0.63              | 0.63              | 0.64              | 0.63              |
| Observations          | 6,399             | 6,399               | 6,399             | 6,399             | 6,399             | 6,399             | 6,399             | 6,399             | 6,399             | 6,399             |

Notes: The first column is a regression of the actual log market-to-book equity on firm characteristics. The remaining columns are regressions of the counterfactual log market-to-book equity on firm characteristics. All characteristics are standardized within each year. Each column corresponds to capital flows from the given group of investors to other institutional investors. All specifications include year fixed effects and indicator variables for a missing environmental score or governance index. The  $t$ -statistics, based on standard errors clustered by firm, are reported in parentheses. The sample period is 2010–19.

### Read:

- The first column repeats the actual valuation regression.
- Each remaining column shows the counterfactual valuation relation after capital flows from one investor type to other institutional investors.
- Small-active advisor outflows raise the environmental valuation coefficient from 0.17 to 0.21.
- Foreign-investor outflows lower the environmental valuation coefficient from 0.17 to 0.14.
- Small-active advisors and hedge funds are influential for the governance coefficient.

**Economic message:** The asset demand system turns narratives like “foreign investors price green firms” or “hedge funds price governance” into counterfactual valuation regressions. The same numbers can be translated into expected-return effects by multiplying valuation changes by approximately -0.097.

## 6 Short summary

- The paper estimates investor-level asset demand curves using holdings and firm characteristics.
- Investor demand is highly heterogeneous, even within investor type.
- Demand is not very price elastic on average; this low elasticity makes capital-flow counterfactuals quantitatively important.
- The active-to-passive shift changes prices substantially, with about 14% equity repricing, but does not meaningfully reduce price informativeness.
- The reason is that capital did not flow systematically from investors with more informative demand shifters to investors with less informative demand shifters.
- Broad climate-induced shifts in demand can create large green-versus-brown valuation effects and redistribute wealth across institutional investors.
- Hedge funds matter disproportionately per dollar because their demand is distinctive and relatively elastic.

## **7 Conclusion**

### **7.1 Core conclusion**

Investor identity matters for asset prices. Prices are determined by an equilibrium of heterogeneous demand curves, not by a single representative investor. Therefore, market trends and policies can be evaluated by asking how they change investor wealth weights or investor demand loadings.

### **7.2 Economic intuition**

A capital flow changes prices when the investor receiving capital wants a different portfolio from the investor losing capital. A demand shift changes prices when investors collectively demand more of a characteristic, such as environmental quality. The resulting price changes then redistribute wealth depending on who already owns the repriced assets.

### **7.3 Takeaway**

The paper provides a practical framework for financial-market counterfactuals. It can quantify not only whether prices move, but also which investor groups caused the move, whether prices become more informative, and who gains or loses from the repricing.